BAYESIAN INFERENCE

Full-Syllabus Critical Thinking Practice Set

135 challenging application-based problems

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7 chapters × 45 topics × 3 questions per topic

Purpose. This question bank is designed to develop derivation skills, model-building judgement, mathematical reasoning, and interpretation across the complete seven-chapter syllabus in the supplied `try.pdf`.

Coverage. Every numbered topic from Sections 1.1 through 7.7 is represented separately. The chapter order and topic names follow the source material.

Solutions. Deliberately omitted. Use the questions for independent practice before consulting notes or worked examples.

Bayesian Inference – Practice Edition
2026–2027

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Instructions for practice

Attempt each problem by first identifying the sampling model, unknown parameter(s), likelihood kernel, prior, and inferential target. Show intermediate algebra, state parameterisation conventions (especially Gamma shape-rate versus scale and variance versus precision), and interpret the final result in the context of the problem. For computational or diagnostic questions, state what the computation estimates and what evidence would make the output trustworthy.

The questions are intentionally multi-step. A correct final number without a justified model, derivation, or interpretation is not a complete solution. When a question asks for a comparison, identify precisely which assumption or inferential operation changes.

Notation convention. Unless a question states otherwise, Gamma(a, b) uses shape-rate parameterisation, with mean a/b and variance a/b^2. The inverse-Gamma kernel is proportional to $x^{-a-1}\exp(-b/x)$.

1 Chapter 1: The Bayesian framework: probability, models, and updating

The problems in this chapter follow the corresponding topics in the supplied syllabus notes. Each topic has three independent questions designed to test both derivation and application.

1.1 What Bayesian inference is

Problem 1. A screening programme is introduced in two districts. In District A the disease prevalence is believed to be 1%, while in District B it is believed to be 8%. The same test has sensitivity 0.96 and specificity 0.93 in both districts. A report gives only the overall positive-test rate and then claims that a positive result has the same meaning in both districts. Construct the full Bayesian model for each district, derive the two posterior disease probabilities after a positive result, and explain precisely why the test characteristics alone do not determine these posteriors. State which quantities are prior information, likelihood information, and posterior information.

Problem 2. A machine has an unknown probability p of producing a defective item. Before observing the current batch, an engineer assigns a distribution to p, observes 12 defects in 200 items, and then predicts the number of defects in the next 50 items. A colleague says that the posterior distribution is simply the probability distribution of the next batch. Write the complete sequence of probability statements needed to distinguish the unknown parameter, observed data, likelihood, prior, posterior, and future data. Explain what additional randomness enters the prediction beyond uncertainty about p.

Problem 3. Two analysts use the same sampling model and observe the same data but begin with different prior distributions. One reports a posterior concentrated near a small parameter value; the other reports a posterior concentrated near a larger value. Give a mathematically coherent example in which this can happen. Identify the circumstances under which the data will eventually dominate the prior, and construct a counterexample or limiting argument showing why a large sample does not automatically rescue a badly misspecified sampling model.

1.2 The three steps of Bayesian data analysis

Problem 4. A public-health team models the number of cases in each district using independent Poisson distributions with a common rate per person-year. The observed counts are far more variable across districts than the fitted model predicts. Lay out the three stages of Bayesian data analysis for this investigation: model construction, posterior computation, and model evaluation. Propose at least three posterior predictive checks that would expose the failure, explain what each check diagnoses, and describe a principled model revision without treating a larger sample or more MCMC iterations as a cure for misspecification.

Problem 5. A laboratory measures patients in four hospitals. An analyst assumes all observations are conditionally independent and identically distributed given one common mean, but hospital-level plots show persistent shifts. Specify two competing Bayesian models, one with a common parameter and one with hospital-specific parameters. For each model, describe the prior specification, the posterior target, and checks you would perform before interpreting the result. Explain how the three stages of analysis interact rather than functioning as a one-way pipeline.

Problem 6. A posterior distribution has been computed exactly under a convenient conjugate model and is mathematically proper. The posterior predictive distribution nevertheless assigns negligible probability to a pattern that appears repeatedly in the observed data. Explain why posterior propriety and model adequacy are logically distinct. Design a model-checking procedure using replicated data, identify what would count as a serious discrepancy, and state how the analysis should change if the discrepancy is systematic rather than an isolated random fluctuation.

1.3 Parameters, data, likelihood, and predictive quantities

Problem 7. Suppose counts satisfy $Y_i \mid \lambda \sim \text{Poisson}(x_i\lambda)$, where the known exposures are $(1, 2, 4, 3)$ and the prior is $\lambda \sim \text{Gamma}(a, b)$ in the shape-rate convention. The observed counts are $(0, 3, 5, 2)$. Derive the likelihood and identify the data summaries through which it depends on λ. Obtain the posterior and derive the posterior predictive distribution for a new count with exposure 2. Explain why substituting the posterior mean of λ into a Poisson distribution does not generally give the correct predictive distribution.

Problem 8. Let $Y_1, \ldots, Y_n \mid \mu$ be conditionally independent normal observations with known variance, and let μ have a normal prior. A future observation Y_{n+1} is to be predicted. Derive the posterior predictive distribution by integrating over the posterior of μ. Compare its variance with the variance of $Y_{n+1} \mid \mu$, and explain which part represents uncertainty about the unknown mean and which part represents irreducible observation noise. Give a numerical design in which the distinction is practically important.

Problem 9. Let $P \sim \text{Beta}(a, b)$ and, conditional on P, let Y_1, Y_2 be independent Bernoulli observations with success probability P. Derive the marginal probability that both observations are successes and the marginal covariance $\text{Cov}(Y_1, Y_2)$. Explain why conditional independence does not imply marginal independence. Then explain how the answer changes if the two observations have separate independent parameters rather than a shared uncertain parameter.

1.4 Posterior odds and the likelihood ratio

Problem 10. A disease has prior prevalence 0.004. A test has sensitivity 0.98 and false-positive probability 0.025. A patient tests positive. Derive the prior odds, likelihood ratio, posterior odds, and posterior probability of disease. A second test is then performed and is positive. Determine when multiplying the two likelihood ratios is valid, and show how the calculation changes if the tests are conditionally dependent given disease status. Explain why multiplying posterior probabilities directly is invalid.

Problem 11. Two hypotheses concern the failure mechanism of a component: H_0 says a failure probability is 0.02, while H_1 says it is 0.08. Prior model probabilities are 0.7 and 0.3. In 20 independent trials, 4 failures are observed. Calculate the posterior odds and posterior probability of H_1. Then repeat the reasoning after a second independent batch of observations. Explain how the direction of the likelihood ratio determines whether the data favour the numerator or denominator hypothesis.

Problem 12. A hospital must decide whether to administer an additional diagnostic procedure. Let the posterior odds of disease be O, the cost of treating a healthy patient be C_{FP}, and the cost of failing to treat a diseased patient be C_{FN}. Derive the posterior-probability or posterior-odds threshold at which treatment minimises posterior expected loss. Explain how a likelihood ratio from a new test changes the decision, and distinguish evidence updating from the final decision rule.

1.5 Exchangeability and conditional independence

Problem 13. A researcher observes n Bernoulli outcomes that are conditionally independent given a common unknown success probability $P \sim \text{Beta}(a, b)$. Derive the joint marginal probability of an ordered sequence containing s successes, and show that it depends on the sequence only through its success count. Calculate the marginal covariance between two distinct outcomes. Explain in precise terms why the sequence is exchangeable but not generally independent.

Problem 14. A hospital selects 3 patient files uniformly without replacement from a finite collection containing exactly 8 positive and 12 negative cases. Let Y_i indicate whether the i-th selected file is positive. Show that the sequence is exchangeable but that the indicators are dependent. Compare this mechanism with conditionally iid Bernoulli sampling given an unknown probability. Explain why exchangeability is weaker than conditional iid sampling under a parameter model.

Problem 15. Three sensors monitor the same machine. Their outputs are independent conditional on an unobserved machine state Z, but the state is uncertain and has two possible values. Specify a joint model and derive the marginal joint probability of two sensor alarms. Give conditions under which the alarms become positively associated after integrating out Z. Explain how an analyst who assumes marginal independence could overstate the strength of combined evidence.

1.6 Worked examples: understanding Bayes' rule

Problem 16. A rare condition affects 0.5% of a population. A first test has sensitivity 0.9 and specificity 0.97. A confirmatory test has sensitivity 0.95 and specificity 0.99, and the tests are conditionally independent given the true condition status. Derive the probability of disease after (i) a positive first test and (ii) two positive tests. Calculate the probability of two positive results under each true status, and explain why the second positive result does not simply multiply the first posterior probability by 0.95.

Problem 17. A genetic marker can arise from one of three mutually exclusive genetic states G_1, G_2, G_3, with prior probabilities $(0.2, 0.5, 0.3)$. A laboratory observation E has probabilities $(0.8, 0.2, 0.1)$ under the three states. Derive the posterior distribution over the states and the posterior odds for every pair. Now suppose the observation is reported only as E or as a coarsened event $E \cup F$. Explain how the likelihoods and posterior conclusions change under the coarsening.

Problem 18. A classifier is evaluated under two possible population models. Under model M_0, a word is correctly classified with probability 0.85; under model M_1, that probability is 0.95. The prior probability of M_1 is 0.25. In 10 independent classifications, 9 are correct. Calculate the posterior model probability and compare it with the maximum-likelihood choice of model. Explain what the Bayesian calculation adds and why the result is not determined by the larger likelihood alone.

1.7 Common conceptual distinctions

Problem 19. An analyst makes the following claims: (i) the likelihood is a probability distribution over the unknown parameter; (ii) a 95% posterior predictive interval means the parameter lies in it with probability 0.95; (iii) a proper posterior proves that the model fits; and (iv) conditional independence guarantees marginal independence. For each claim, determine whether it is correct, give a counterexample where it is not, and replace it with a mathematically accurate statement. Your answer must distinguish prior, likelihood, posterior, parameter, and future observation.

Problem 20. A posterior distribution for a defect probability is sharply concentrated, but posterior predictive replications almost never reproduce the number of extreme defect clusters observed in the actual factory. Explain how a posterior can be concentrated and proper while the model remains inadequate. Design a posterior predictive discrepancy statistic for clustering and explain what conclusions can and cannot be drawn from the observed discrepancy.

Problem 21. A report gives a narrow 95% credible interval for a population mean and a much wider 95% predictive interval for an individual future measurement. A manager says that the two intervals are inconsistent because they concern the same population. Explain the distinction by decomposing predictive uncertainty into parameter uncertainty and observation-level randomness. Construct a normal example in which the credible interval becomes arbitrarily narrow with increasing sample size while the predictive interval retains a nonzero width.

2 Chapter 2: Exponential families and conjugate prior distributions

The problems in this chapter follow the corresponding topics in the supplied syllabus notes. Each topic has three independent questions designed to test both derivation and application.

2.1 Why exponential families are useful

Problem 22. Write the Bernoulli, Poisson, and normal-with-known-variance likelihoods in the canonical exponential-family form $h(y)\exp\{\eta T(y) - A(\eta)\}$. For each model, identify the natural parameter, sufficient statistic, log-partition function, and base measure. Then show how the joint likelihood of an iid sample reduces to a function of a low-dimensional summary. Explain why this representation is useful for conjugate updating, without claiming that membership in an exponential family makes every prior scientifically appropriate.

Problem 23. Two independent datasets are observed under the same Poisson-rate model. Dataset A contains 10 counts with total 24, and Dataset B contains 40 counts with total 24; exposures differ but have the same total in both datasets. Determine whether the two datasets produce the same likelihood kernel for the rate. Extend the argument to a Gamma prior and identify exactly which summaries must match for the posterior distributions to be identical. Explain what information is discarded by those summaries and why it is irrelevant to this particular parameter under this model.

Problem 24. Consider a normal sample with known variance and unknown mean. Starting from the joint density, isolate the part involving the unknown mean and identify a sufficient statistic. Now suppose one observation is replaced by two values whose sum is unchanged but whose individual values are very different. Determine whether the likelihood for the mean changes. Explain why sufficiency is model- and parameter-specific, and identify which aspects of the data would matter if the variance were also unknown.

2.2 Beta-Bernoulli and Beta-Binomial models

Problem 25. A reliability engineer assigns $P \sim \text{Beta}(2, 5)$ to the probability of successful operation. The first 20 trials give 14 successes. Derive the posterior distribution, posterior mean, and probability that the next 5 trials contain at least 4 successes. Do not treat the future trials as having a fixed success probability equal to the posterior mean: derive the predictive probability by integrating over P, and explain the extra variability created by uncertainty about P.

Problem 26. A prior is to be chosen for a Bernoulli success probability. The expert states that the prior mean must be 0.3 and the prior effective sample size $a + b$ must be 12. Determine the Beta hyperparameters. After observing 2 successes in 5 trials, derive the posterior and compare the posterior mean with the sample proportion. Then discuss whether the pseudo-count interpretation means the prior is literally equivalent to 12 additional observed trials in every inferential respect.

Problem 27. Two laboratories observe the same number of successes but in different batches and in different orders. One analyst updates a Beta prior after each observation; another uses the final success and failure totals in one batch. Prove that the final posterior is the same under the iid Bernoulli model. Then derive the predictive probability of the next two outcomes in the two possible orders (success-failure and failure-success), and explain how exchangeability is reflected in the result.

2.3 Gamma-Poisson model, including exposure

Problem 28. A clinic records counts $Y_i \mid \lambda \sim \text{Poisson}(x_i\lambda)$ with exposures $x = (0.5, 1, 2, 4)$, counts $y = (0, 2, 3, 7)$, and prior $\lambda \sim \text{Gamma}(3, 2)$ in the shape-rate convention. Derive the posterior and calculate its mean and variance. Then derive the posterior predictive distribution for a new observation with exposure 1.5 and calculate its probability of zero events. Explain why using the number of clinics rather than total exposure in the posterior rate is incorrect.

Problem 29. Two regions have observed counts 12 and 18 over exposures 3 and 9, respectively. Under a common rate model with a Gamma prior, derive the posterior. Then consider a model with separate regional rates and independent Gamma priors. Determine which information is lost by imposing a common rate, and describe a posterior predictive comparison that could reveal whether the common-rate assumption is unreasonable. Your answer must distinguish evidence about the common rate from evidence about the adequacy of the common-rate model.

Problem 30. A data analyst reports that the posterior predictive number of cases over exposure 10 is Poisson with mean equal to 10 times the posterior mean of λ. Derive the correct posterior predictive distribution under a Gamma-Poisson model and compare its variance with the plug-in Poisson variance. Identify the parameter regime in which the discrepancy is largest, and explain how posterior uncertainty produces overdispersion relative to a fixed-rate Poisson model.

2.4 Gamma-Exponential model

Problem 31. A reliability study records exact failure times 1.2, 2.1, and 0.7, together with right-censoring times 1.5 and 3.0 for two units that have not failed. Assume an exponential rate λ and prior $\text{Gamma}(a, b)$. Derive the likelihood contribution of each exact and censored observation, obtain the posterior, and explain why the shape increases by the number of observed failures while the rate increases by total time at risk. State what would go wrong if the censored units were incorrectly treated as failures at their censoring times.

Problem 32. Suppose $\lambda \mid y \sim \text{Gamma}(a_n, b_n)$ in the shape-rate convention, but the scientific quantity of interest is the mean waiting time $\tau = 1/\lambda$. Derive the posterior density of τ using the Jacobian. Determine its mode and state the conditions under which its mean exists. Explain why simply replacing λ by $1/\tau$ in the Gamma density gives an incorrect transformed density.

Problem 33. A survival study has three exact event times and four right-censored observations. Two analysts use the same total observed time, but one records the number of failures incorrectly as seven. Derive the correct posterior and compare it with the incorrect one. Explain which part of the likelihood carries information about the event rate and how the event indicator and total exposure play different roles. Give a general expression for the likelihood with event indicators.

2.5 Dirichlet-Multinomial model

Problem 34. A website has four possible landing-page outcomes with probability vector $\boldsymbol{p} \sim \text{Dirichlet}(1, 2, 1, 3)$. In 100 visits, the observed counts are $(10, 20, 15, 55)$. Derive the posterior distribution, posterior mean vector, and posterior predictive probability that the next 10 visits contain exactly 2 outcomes of type 1 and 5 of type 4 (with the remaining 3 visits split between types 2 and 3 in any way). Show how the multinomial coefficient and Dirichlet normalising constants enter the calculation.

Problem 35. Let $\boldsymbol{p} \sim \text{Dirichlet}(\alpha_1, \alpha_2, \alpha_3)$. Researchers decide to combine categories 1 and 2 into a single category after collecting data. Derive the marginal posterior distribution for $p_1 + p_2$, and compare it with the posterior obtained by aggregating the data before updating. Explain why the Beta marginal property is useful and state the conditions under which the result holds.

Problem 36. For a three-category multinomial model with a Dirichlet prior, derive the covariance between two distinct category probabilities under the posterior. Explain why the components of the probability vector are negatively dependent even though the data are conditionally iid given the vector. Then derive the predictive probability that the next two observations belong to different categories and compare it with the product of the one-step predictive probabilities.

2.6 Normal likelihood with a conjugate Normal prior: known variance

Problem 37. A sensor reports $Y_i \mid \mu \sim N(\mu, 9)$, with prior $\mu \sim N(10, 4)$. A sample of size 5 has mean 13. Derive the posterior distribution by completing the square and identify the weights assigned to the prior mean and sample mean. Determine the smallest sample size, with sample mean fixed at 13, for which the posterior mean lies within 0.5 of 13. Interpret this as a competition between prior precision and data precision.

Problem 38. An experiment is performed in two stages. The first stage has sample size 8 and mean 4.2; the second has sample size 12 and mean 5.1. The observation variance is known to be 4 and the prior on the unknown mean is normal. Derive the posterior after stage one and then update it using stage two. Prove that the final posterior is identical to the posterior obtained by analysing all 20 observations at once. Explain which summary is sufficient for sequential updating.

Problem 39. A scientist claims that a normal prior $N(\mu_0, \tau_0^2)$ is 'weak' because its variance is numerically large. The measurement unit is then changed from metres to millimetres. Show how the prior parameters and likelihood variance transform, and demonstrate that the posterior for the physical mean is unchanged when the transformation is handled correctly. Explain why judging prior informativeness from a bare numerical variance without reference to the measurement scale is misleading.

2.7 Normal-Gamma and Normal-Inverse-Gamma conjugacy

Problem 40. Let $Y_1, \ldots, Y_n \mid \mu, \sigma^2$ be iid normal, with $\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0)$ and $\sigma^2 \sim \text{InvGamma}(a_0, b_0)$. Derive the joint posterior and all updated hyperparameters in terms of n, $\bar{Y}$, and $S = \sum_i (Y_i - \bar{Y})^2$. Your derivation must show where the between-centre term involving $(\bar{Y} - \mu_0)^2$ enters and explain its role in the variance update.

Problem 41. A statistician uses a Normal-Gamma prior on precision $\tau = 1/\sigma^2$, while a colleague writes an apparently similar Normal-Inverse-Gamma prior on variance. Translate one parameterisation into the other, including the density transformation. Identify the Gamma shape-rate convention and the inverse-Gamma kernel explicitly. Explain how a missing Jacobian or a confusion between rate and scale would alter the posterior.

Problem 42. Under a Normal-Inverse-Gamma prior, derive the posterior predictive distribution of one future observation after integrating out both the regression mean and variance. State its location, scale, and degrees of freedom in terms of the updated hyperparameters. Explain why the predictive distribution has heavier tails than a normal distribution with the posterior mean and posterior expected variance plugged in.

2.8 Exponential-family summary table

Problem 43. A study contains four data streams: (i) binary treatment responses, (ii) event counts observed over unequal person-time exposures, (iii) exact and right-censored waiting times, and (iv) counts in five mutually exclusive outcome categories. For each stream, identify a likelihood family and a conjugate prior from the chapter. Derive the likelihood kernel and posterior update symbolically, carefully stating which data summary updates each hyperparameter. Explain why the same 'add the observations to the prior' slogan is not adequate for all four models.

Problem 44. A researcher has access only to the following summaries: number of successes and failures; total count and total exposure; number of exact events and total time at risk; and category counts. For each summary, determine whether it is sufficient for the corresponding unknown parameter under the model in the notes. Construct two raw datasets with identical sufficient summaries but visibly different observation patterns, and explain why they nevertheless yield the same posterior under the specified model. State one plausible model extension under which the discarded pattern could become relevant.

Problem 45. A company wants to predict a future batch of binary outcomes, a future count over a known exposure, a future waiting time, and the next categorical outcome. For each case, derive or identify the posterior predictive distribution implied by the conjugate model. Compare the source of predictive uncertainty and explain why a plug-in estimate generally understates uncertainty. Your answer should connect the likelihood kernel, prior kernel, posterior family, and predictive family rather than merely listing distribution names.

3 Chapter 3: Bayesian estimation, MAP, credible intervals, and prior choice

The problems in this chapter follow the corresponding topics in the supplied syllabus notes. Each topic has three independent questions designed to test both derivation and application.

3.1 Posterior distributions contain more than a point estimate

Problem 46. A posterior distribution for a treatment effect is a mixture: with probability 0.65 it is $N(0.2, 0.1^2)$, and with probability 0.35 it is $N(1.1, 0.25^2)$. Determine the posterior mean and identify the posterior median and mode conceptually (numerical computation may be used for the median and mode). For squared-error, absolute-error, and zero-one neighbourhood loss, explain which summary is the relevant Bayes estimator. Discuss why reporting only the posterior mean can conceal a scientifically important second mode.

Problem 47. Let $\theta > 0$ have a posterior density that is Gamma with shape 3 and rate 2. Find the posterior mode of θ, and then transform to $\phi = \log \theta$. Derive the density of ϕ and find its mode. Explain why the transformed mode is not generally the logarithm of the original mode, and state which posterior summaries are invariant under a one-to-one transformation and which are not.

Problem 48. A decision maker must choose an action a and incurs loss $L(a, \theta) = (a - \theta)^2 + 4|a - \theta|$. Derive the posterior expected loss and characterise the Bayes action. Compare it with the posterior mean and median. Explain why the optimal point estimate is a property of both the posterior distribution and the loss function, rather than a universal feature of the posterior.

3.2 Posterior variance and the value of data

Problem 49. Let θ have prior variance 16. A future observation Y can take values 0 or 1, and the posterior means are known to be 2 when $Y = 0$ and 6 when $Y = 1$, with prior predictive probabilities 0.75 and 0.25. Use the law of total expectation and law of total variance to determine the prior mean and the expected posterior variance. Explain the role of $\mathrm{Var}\{E(\theta \mid Y)\}$ as the amount of prior uncertainty resolved by the data.

Problem 50. A very concentrated prior is centred at 0, but the observed data strongly favour values near 5. Construct a conjugate normal-normal example in which the posterior variance for this particular dataset is larger than the prior variance, or prove that such an example is impossible under the specified known-variance normal model. Reconcile your conclusion with the law of total variance. Your answer must distinguish a statement about every realised dataset from a statement averaged over the prior predictive distribution.

Problem 51. A laboratory can collect either 10 additional measurements from an existing group or 10 measurements from a new group. Formulate a Bayesian design argument using expected posterior variance to compare the two plans in a hierarchical setting. Identify which unknown quantity each plan is expected to inform and explain why the value of data cannot be assessed solely by counting the number of additional observations.

3.3 Credible intervals

Problem 52. After observing data, a parameter has posterior distribution Beta$(2, 8)$. Define the equal-tailed 90% credible interval and the 90% highest posterior density (HPD) interval. Explain how each is constructed and why they need not have the same endpoints. Without relying on a normal approximation, describe how you would compute both intervals numerically and verify their posterior probability content.

Problem 53. A posterior density is bimodal, with 60% of its probability near -2 and 40% near 3, and very low density around zero. Construct a 90% central credible interval and describe the possible shape of a 90% HPD region. Explain why a single interval may include a low-density region and why the HPD set may be disconnected. Discuss what each representation communicates to a scientist.

Problem 54. Let θ have a continuous posterior and let $\phi = \exp(\theta)$. A 95% equal-tailed interval for θ is $[l, u]$. Derive the corresponding interval for ϕ, and explain why it remains an equal-tailed interval. Then explain why transforming a 95% HPD interval for θ does not generally produce the 95% HPD interval for ϕ. Identify the role of the Jacobian in this difference.

3.4 Informative, weakly informative, and noninformative priors

Problem 55. An expert believes a failure probability is usually below 0.05 but cannot rule out values as high as 0.2. Propose two proper prior distributions that express different degrees of information while respecting this belief. Compare their prior predictive implications for a batch of 20 trials. Explain why a prior can be weakly informative on the probability scale but highly informative on the log-odds scale, and state how you would communicate the elicitation assumptions.

Problem 56. A colleague proposes a flat prior on a positive rate λ and calls it noninformative. Another proposes a flat prior on the mean waiting time $\tau = 1/\lambda$. Derive the induced density under each parameterisation and show whether the two priors express the same beliefs. Discuss propriety and explain why 'flat' is not a parameterisation-free concept.

Problem 57. Two reasonable priors lead to posterior means 0.04 and 0.07 for a risk parameter, while the posterior probability that risk exceeds 0.1 is 0.02 under one prior and 0.18 under the other. Design a prior-sensitivity analysis that goes beyond comparing posterior means. Specify which prior predictive checks, posterior summaries, and scientific conclusions should be compared, and explain what result would make you describe the conclusion as prior-sensitive.

3.5 Jeffreys' prior

Problem 58. For $X \sim \text{Binomial}(n, p)$, reparameterise using $\eta = \log\{p/(1-p)\}$. Write the likelihood in terms of η, derive the Fisher information for η, and obtain Jeffreys' prior directly on the log-odds scale. Transform it back to the probability scale and explain how the Jacobian reconciles the two expressions. Compare the result with a prior that is uniform in p.

Problem 59. Let $Y_1, \ldots, Y_n$ be iid exponential observations with unknown rate $\lambda > 0$. Derive the Fisher information for λ and Jeffreys' prior. Then transform to the mean waiting time $\tau = 1/\lambda$ and derive the induced prior. Explain how this example illustrates the invariance motivation for Jeffreys' construction, and discuss whether the resulting prior is proper.

Problem 60. A model has density $f(y \mid \theta)$ and Fisher information $I(\theta)$. A student computes $I(\theta)$ from the observed sample rather than taking its expectation, then uses its square root as a prior. Explain why this is not the standard Jeffreys prior described in the notes. Give a complete derivation template for the scalar case and discuss at least two cautions that remain even after Jeffreys' prior has been derived correctly.

3.6 Prior sensitivity and transformations

Problem 61. Let p have a Beta$(2, 3)$ prior and define the odds $\omega = p/(1-p)$. Derive the density of ω with the correct Jacobian and identify its support. Compare this induced prior with a prior that is uniform on ω. Explain why choosing a prior before deciding which parameter scale has substantive meaning can change the analysis.

Problem 62. A scientific conclusion is stated as $\Pr(\theta > c \mid y) > 0.95$. Repeat the analysis under three reasonable priors that differ in concentration and tail behaviour. Design a sensitivity table that records posterior means, interval endpoints, and the probability of the scientific event. Explain why sensitivity should focus on the decision-relevant conclusion rather than searching for the prior that produces a preferred answer.

Problem 63. An analyst reports that the posterior mode and the 95% HPD interval are unchanged after replacing a parameter by a nonlinear one-to-one transformation. Determine whether these invariance claims are generally valid. Use a concrete transformation to show the effect of the Jacobian on the density and explain which probability statements are preserved even when modes and HPD sets change.

4 Chapter 4: Bayesian hypothesis testing and Bayes factors

The problems in this chapter follow the corresponding topics in the supplied syllabus notes. Each topic has three independent questions designed to test both derivation and application.

4.1 Simple and composite hypotheses

Problem 64. A coin is tested under $H_0 : p = 0.5$ and $H_1 : p$ unknown with $p \sim \text{Beta}(2, 2)$. In 12 tosses, 10 heads are observed. Derive the likelihood under H_0, the marginal likelihood under H_1, and the Bayes factor BF_{10}. Explain why evaluating the alternative likelihood at its maximum-likelihood estimate is not the same as computing its marginal likelihood.

Problem 65. A measurement Y is normal with known variance 1. Under H_0, the mean is fixed at zero; under H_1, the mean has prior $N(0, \tau^2)$. For an observed sample mean $\bar{y}$ from n measurements, derive the two marginal likelihoods and the Bayes factor. Identify the integral over the alternative parameter and explain which model quantities must be specified before posterior model probabilities can be reported.

Problem 66. Two candidate models describe the same count data. Model A fixes the event rate at a specified value. Model B assigns a Gamma prior to an unknown event rate. Formulate the marginal likelihood under each model and derive the Bayes factor symbolically. Explain how the prior distribution under Model B affects its evidence even when Model B can fit the observed count better at its maximum-likelihood estimate.

4.2 Interpreting a Bayes factor

Problem 67. A study reports $BF_{10} = 12$, but the prior probability of H_1 is only 0.01. Calculate the posterior probability of H_1. A second analyst incorrectly reports 12% as the posterior probability. Explain the error and determine what Bayes factor would be required to make the posterior probability of H_1 exceed 0.5 under the same prior model probability.

Problem 68. Three independent research teams collect evidence y_1, y_2, y_3 under the same two fully specified hypotheses. Show when the combined Bayes factor equals the product of the three study-specific Bayes factors. Construct a situation in which this multiplication fails because the studies share an unknown nuisance quantity or their data are conditionally dependent. Explain what joint model is needed in that case.

Problem 69. A Bayes factor of 0.08 is reported for H_1 against H_0. One report calls this strong evidence for H_1, while another calls it evidence for H_0. Resolve the disagreement by fixing the numerator and denominator convention, convert the factor to the reciprocal comparison, and show how prior odds determine the final posterior odds.

4.3 Point null hypotheses and continuous alternatives

Problem 70. Let $Y_1, \ldots, Y_n \mid \theta \sim N(\theta, \sigma^2)$ with known σ^2. Compare $H_0 : \theta = 0$ against $H_1 : \theta \sim N(0, \tau^2)$. Derive the Bayes factor in terms of $n, \bar{y}, \sigma^2, \tau^2$. Analyse its behaviour as $\tau^2 \to 0$ and $\tau^2 \to \infty$, and explain why a very diffuse alternative can receive a low marginal likelihood even when it contains values that fit the data extremely well.

Problem 71. For binomial data $Y \sim \text{Binomial}(n, p)$, compare the point null $H_0 : p = 0.4$ with a continuous alternative $H_1 : p \sim \text{Beta}(a, b)$. Derive the Bayes factor after observing y successes. Explain why the null receives positive prior probability only because it is represented as a separate model, and why a continuous Beta prior alone assigns probability zero to the event $p = 0.4$.

Problem 72. A very small p-value is obtained for testing $H_0 : \theta = 0$, yet the Bayes factor favours the null under a diffuse alternative. Construct a coherent normal-model explanation of how this can occur. Identify the role of sample size, effect size, prior scale, and the point mass at the null. Explain why a p-value is not the posterior probability of the null and why the two procedures answer different questions.

4.4 Sensitivity, improper priors, and limitations

Problem 73. Suppose a model comparison uses an improper prior $\pi(\theta) = c\, g(\theta)$, where the arbitrary constant c is unknown. Show how the constant enters the marginal likelihood and explain why the Bayes factor may depend on an arbitrary choice. Give an example of why posterior propriety for parameter estimation does not automatically make an improper prior valid for model comparison. State what must be done before a Bayes factor can be interpreted.

Problem 74. For a composite alternative, calculate or analyse the Bayes factor under a family of prior scales $\tau \in \{0.1, 1, 10, 100\}$. Design a sensitivity analysis that distinguishes changes caused by the data from changes caused by the prior. Explain the role of prior predictive mass near the observed effect and why a single reported Bayes factor can create a misleading impression of prior-independent evidence.

Problem 75. Two independent datasets are analysed sequentially. A colleague claims that $BF_{10}(y_1, y_2) = BF_{10}(y_1) BF_{10}(y_2)$ always holds. State the conditions under which the identity is valid, derive the identity from joint marginal likelihoods, and give a counterexample involving a shared latent parameter or dependence between batches. Explain how the correct joint analysis avoids double-counting evidence.

5 Chapter 5: Posterior simulation: Gibbs sampling, Metropolis-Hastings, and diagnostics

The problems in this chapter follow the corresponding topics in the supplied syllabus notes. Each topic has three independent questions designed to test both derivation and application.

5.1 Why MCMC is required

Problem 76. A logistic regression model has several coefficients with correlated posterior dependence and no closed-form normalising constant. Specify the posterior target and explain why direct sampling and exact integration may be impractical. Describe how MCMC draws can approximate a posterior mean, a tail probability, and a posterior predictive quantity. State the conditions that make the simulation output meaningful and explain why a long run does not by itself establish model correctness or convergence.

Problem 77. Consider a posterior density known only up to proportionality, $\pi(\theta \mid y) \propto q(\theta)$, where direct sampling is unavailable but pointwise evaluation of q is possible. Explain why this is a natural setting for MCMC. Compare the computational target with numerical quadrature and independent Monte Carlo, and identify which features of a Markov chain are deliberately used to produce draws from the target distribution.

Problem 78. A posterior has two separated modes and a sampler is started close to the first mode. After one million iterations, the trace appears stable and the posterior mean is reported. Explain why the run may still be inadequate. Design a multiple-chain strategy with dispersed initial values and propose diagnostics that would reveal failure to explore the second mode. Discuss how a failure of exploration affects posterior means and tail probabilities.

5.2 Gibbs sampling

Problem 79. Let (X, Y) have a bivariate normal target distribution with zero means, unit variances, and correlation ρ. Derive the two full conditional distributions. Starting from a specified state $(x^{(0)}, y^{(0)})$, write one complete Gibbs sweep and explain the order in which the newly updated and previous values are used. Analyse qualitatively how the chain's mixing changes as $|\rho|$ approaches one.

Problem 80. Suppose $Y_i \mid \theta_i \sim N(\theta_i, s_i^2)$, $\theta_i \mid \mu, \tau^2 \sim N(\mu, \tau^2)$, and $\mu \sim N(m_0, v_0)$, with τ^2 known. Derive the full conditional distributions of each θ_i and of μ. Write a Gibbs algorithm for one iteration and explain why each update conditions on the most recently available values of the other blocks.

Problem 81. A proposed Gibbs sampler uses the correct full conditional for θ_1 but accidentally conditions on the old value of θ_1 when updating θ_2, rather than the newly drawn value required by the specified scan. Analyse whether this changes the transition mechanism and what must be checked to establish the intended joint posterior as stationary. Explain why having individually recognisable conditional distributions is not, by itself, a complete proof that a proposed implementation is correct.

5.3 Metropolis-Hastings

Problem 82. A Metropolis-Hastings chain targets an unnormalised density $q(\theta)$. From current state $\theta = 1$, it proposes $\theta^* = 2$, with $q(1) = 0.10$, $q(2) = 0.06$, forward proposal density $g(2 \mid 1) = 0.4$, and reverse proposal density $g(1 \mid 2) = 0.2$. Calculate the acceptance probability. Repeat the calculation for a symmetric proposal and explain why omitting the proposal ratio in the asymmetric case produces the wrong stationary distribution.

Problem 83. A random-walk Metropolis algorithm proposes a point with lower posterior density than the current point. A student insists that the proposal must be rejected. Use the acceptance rule to show why this is false. Explain the role of occasional downhill moves in exploring the target distribution, and describe what happens to the chain state after rejection. Include the resulting effect on the sequence of dependent draws.

Problem 84. A Metropolis-Hastings algorithm has an acceptance rate of 0.02 with very large proposals. After reducing the proposal scale, its acceptance rate rises to 0.95, but the trace plot moves only slowly across the parameter space. Explain why neither extreme acceptance rate guarantees good mixing. Propose a tuning and diagnostic strategy that assesses movement, autocorrelation, and exploration of the target rather than optimising acceptance rate alone.

5.4 Convergence: what must be checked

Problem 85. Three chains are run from dispersed initial values. The trace plots show: chain 1 fluctuates around 0; chain 2 fluctuates around 4; chain 3 drifts slowly from 5 toward 2. The marginal density plots are visibly different. Diagnose the evidence and explain why pooling the draws would be misleading. Specify what additional graphical and numerical checks you would perform before deciding whether the chains have explored a common target distribution.

Problem 86. For a multivariate posterior, the trace plot of each coefficient appears stationary, but a derived quantity $\beta_1 + \beta_2$ shows long-range drift and high autocorrelation. Explain why diagnostics must include derived quantities and not only the original parameters. Design a set of diagnostics for the quantity of scientific interest and state what patterns would make you withhold posterior summaries.

Problem 87. A sampler appears to converge for the posterior mean but not for a 0.99 posterior quantile. Explain why tail functionals can be more difficult to estimate than central summaries. Describe how trace plots, autocorrelation plots, multiple chains, and Monte Carlo standard errors would be used to assess the tail probability. Explain why a single convergence statistic cannot certify all inferential quantities simultaneously.

5.5 Scale reduction and effective sample size

Problem 88. Four chains each retain 5,000 draws. Their lag autocorrelations are approximately $\rho_k = 0.7^k$. Approximate the integrated autocorrelation time and effective sample size for one chain, then for the combined retained draws under comparable mixing. Estimate the Monte Carlo standard error of a posterior mean when the posterior standard deviation is 2. Explain why 20,000 saved draws do not necessarily represent 20,000 independent observations.

Problem 89. Two chains have similar within-chain variances but substantially different chain means. Explain how the between-chain and within-chain variation enter the scale-reduction diagnostic. A reported $\widehat{R} = 1.01$ is presented as proof of convergence. Critique this conclusion and identify the graphical checks and model-specific questions that should accompany the diagnostic.

Problem 90. A posterior mean is estimated as 0.42 with posterior standard deviation 0.15 and effective sample size 225. Calculate the approximate Monte Carlo standard error. The analyst reports the interval $0.42 \pm 1.96(0.15)$ as a Monte Carlo error interval. Explain the conceptual error, calculate the appropriate Monte Carlo precision scale, and distinguish uncertainty about the parameter from numerical error in estimating a posterior summary.

5.6 Warm-up, thinning, and practical reporting

Problem 91. A researcher discards the first 1,000 iterations of a chain and claims that convergence is thereby guaranteed. Explain why warm-up removal is not a convergence test. Propose a defensible procedure for choosing warm-up length using trace plots, multiple chains, and numerical diagnostics. Explain how the decision should be revisited if different parameters or derived quantities show different convergence behaviour.

Problem 92. A chain produces 100,000 highly autocorrelated draws. One analyst retains every 100th draw and claims that thinning has created 1,000 independent posterior draws. Evaluate the claim. Explain what thinning does and does not accomplish, and compare it with retaining all post-warm-up draws while calculating an effective sample size and Monte Carlo standard error.

Problem 93. Write a reproducibility report template for an MCMC analysis of a hierarchical model. It must include the sampler, target distribution, number of chains, starting values, warm-up, retained iterations, thinning decision, trace and autocorrelation checks, scale-reduction diagnostic, effective sample sizes, and Monte Carlo standard errors. Explain why each item helps a reader assess the reliability of the posterior summaries.

6 Chapter 6: Bayesian linear models, ridge, LASSO, and spike-and-slab

The problems in this chapter follow the corresponding topics in the supplied syllabus notes. Each topic has three independent questions designed to test both derivation and application.

6.1 The Gaussian linear regression model

Problem 94. Let $y \mid \beta, \sigma^2 \sim N_n(X\beta, \sigma^2 I_n)$. Derive the likelihood as a function of β, state the dimensions of every matrix, and obtain the posterior kernel under a flat prior on β when X has full column rank. Explain what changes when X is rank deficient and why the likelihood alone cannot identify coefficient combinations in the null space of X.

Problem 95. A regression has 30 observations and 40 predictors. The design matrix has rank 28. Explain why the ordinary least-squares coefficient vector is not uniquely identified. Show how a proper Gaussian prior on β can yield a proper posterior, and discuss which aspects of the result then come from the data and which are regularised by the prior. Do not confuse a numerically stable estimate with evidence that every coefficient is identified by the sample.

Problem 96. Two design matrices X_1 and X_2 have the same dimensions but different predictor scaling. A colleague uses the same independent normal prior variance for every coefficient in both parameterisations. Explain why this may imply different prior beliefs about fitted effects. Construct a rescaling argument and state how the coefficient prior should transform to preserve the same prior distribution over the linear predictor.

6.2 Normal-Inverse-Gamma prior and posterior

Problem 97. Assume $\beta \mid \sigma^2 \sim N_p(b_0, \sigma^2 V_0)$ and $\sigma^2 \sim \text{InvGamma}(a_0, d_0)$. Starting from the Gaussian regression likelihood, derive the posterior quantities V_n, b_n, a_n, and d_n by completing the matrix square. Show every matrix dimension and identify the residual and prior quadratic forms. Explain why simply quoting the updated hyperparameters is insufficient evidence of a correct derivation.

Problem 98. In a regression problem, the prior mean b_0 is far from the least-squares estimate. Derive the posterior inverse-Gamma scale and identify the term measuring disagreement between the prior centre and the data-supported coefficient vector. Explain how this term affects posterior uncertainty about σ^2, and analyse what happens when the prior covariance becomes very large or very small.

Problem 99. Two analysts use the same data but one works with centred predictors and the other with uncentred predictors. Derive the transformation of the coefficient vector and explain how a Normal-Inverse-Gamma prior must be transformed to represent the same prior over the regression function. Discuss how failing to transform the prior can produce different posterior conclusions even though the likelihood models describe the same fitted values.

6.3 Posterior prediction in linear regression

Problem 100. For a new predictor vector x_*, derive the posterior predictive distribution of a future response under the Normal-Inverse-Gamma regression model. Integrate over both β and σ^2, and state the resulting degrees of freedom, location, and scale. Compare the predictive uncertainty for an individual response with uncertainty about the conditional mean $x_*^\top \beta$. Explain why the individual prediction distribution contains an additional observation-noise contribution.

Problem 101. A new observation is to be predicted at a predictor vector x_* with very high leverage, so that $x_*^\top V_n x_*$ is much larger than for points near the centre of the training design. Use the posterior predictive variance to explain why the prediction interval widens. Discuss why reporting only the fitted mean or posterior mean coefficient vector hides this uncertainty.

Problem 102. A fitted regression is used to predict far outside the range of observed predictor values. The posterior predictive interval is computed correctly under the assumed linear model but is still narrow. Explain why correct propagation of parameter uncertainty does not guarantee reliable extrapolation. Identify the modelling assumption being extended beyond the data and propose posterior predictive or prior predictive checks that could reveal implausible extrapolation behaviour.

6.4 Ridge regression as a Gaussian-prior MAP estimate

Problem 103. Assume a Gaussian regression likelihood and independent priors $\beta_j \sim N(0, \tau^2)$. Derive the negative log posterior and show that its minimiser is the ridge estimator. Express the ridge penalty in terms of σ^2 and τ^2. Explain how changing the prior variance changes shrinkage and why the intercept is often treated separately. Then discuss what is lost when only the MAP estimate is reported instead of the full posterior.

Problem 104. Two predictors are highly correlated and both have large ordinary least-squares coefficients with opposite signs. Explain how a ridge prior changes the optimisation problem and why it can stabilise the coefficient estimates. Analyse the limiting behaviour as the prior variance tends to zero and infinity. Explain why ridge usually shrinks coefficients continuously rather than setting them exactly to zero.

Problem 105. A researcher standardises predictors, fits ridge regression, and then converts coefficients back to their original units. Another researcher fits ridge without standardising and uses the same numerical penalty. Explain why these procedures generally impose different prior information. Derive the effect of rescaling a predictor on the coefficient prior and discuss how prior scales should be chosen when predictors have different units.

6.5 LASSO as a Laplace-prior MAP estimate

Problem 106. Assume independent Laplace priors $p(\beta_j) = (\lambda/2)\exp(-\lambda|\beta_j|)$ in a Gaussian regression model. Derive the negative log posterior and show that its MAP estimate minimises a least-squares criterion plus an L_1 penalty. In the one-dimensional orthogonal case, derive the soft-thresholding rule and identify the threshold below which the MAP estimate is exactly zero.

Problem 107. A student applies the scalar soft-thresholding formula separately to two highly correlated predictors and claims to have solved the multivariable LASSO problem. Explain why the coordinate-wise rule does not generally solve the correlated multivariable problem in one independent step. Discuss how correlation can make variable selection unstable and why the optimisation must account for the joint design matrix.

Problem 108. A continuous Laplace prior is said to assign positive posterior probability to the event $\beta_j = 0$ because the LASSO MAP estimate is sometimes exactly zero. Determine whether this is correct. Contrast the density at zero with a point mass at zero, and explain why an exact posterior inclusion probability requires a spike component rather than an ordinary continuous Laplace prior.

6.6 Spike-and-slab priors and Bayesian variable selection

Problem 109. For one coefficient, let $\gamma \sim \text{Bernoulli}(q)$, with $\beta = 0$ when $\gamma = 0$ and $\beta \sim N(0, \tau^2)$ when $\gamma = 1$. Derive the posterior inclusion odds in terms of the prior inclusion odds and the Bayes factor comparing inclusion with exclusion. Explain how the posterior inclusion probability differs from a continuous shrinkage estimate of the coefficient.

Problem 110. Two predictors are strongly correlated and each has a high marginal association with the response. Under a spike-and-slab prior, explain why the posterior inclusion probabilities may be sensitive to whether the predictors are considered jointly or separately. Formulate the model indicators for the two predictors, identify the four possible models, and discuss why model averaging can be more informative than reporting only the single most probable model.

Problem 111. A model has 20 candidate predictors and independent prior inclusion probability $q = 0.1$ for each. Calculate the prior expected number of included predictors and the prior probability of the null model. Explain how the prior inclusion probability affects multiplicity and model uncertainty. Discuss the computational challenge of exploring 2^{20} possible models and how posterior inclusion probabilities summarise uncertainty across them.

6.7 Ridge, LASSO, and spike-and-slab compared

Problem 112. A prediction team wants stable predictions from highly correlated predictors but does not need a sparse model. A scientific team wants an interpretable sparse point estimate. A third team wants posterior probabilities that each predictor belongs in the model. For each objective, choose among ridge, LASSO, and spike-and-slab, justify the prior structure, and state what uncertainty the selected approach preserves or discards. Explain why the three procedures answer related but non-identical questions.

Problem 113. Consider one orthogonalised predictor with standardised score z. Compare the MAP estimates under a zero-centred Gaussian prior and a Laplace prior. Derive the continuous shrinkage rule for ridge and the thresholding rule for LASSO. Then explain how a spike-and-slab prior adds a discrete inclusion decision that neither continuous prior provides. Identify situations in which the three estimates may coincide and situations in which they differ sharply.

Problem 114. A report compares ridge, LASSO, and spike-and-slab solely by the number of zero coefficients. Critique this comparison. Propose a fair evaluation that distinguishes predictive stability, sparse point estimation, posterior inclusion uncertainty, and model averaging. State what should be reported for each method and explain why a common numerical penalty value does not make the prior assumptions comparable.

7 Chapter 7: Hierarchical Bayes, empirical Bayes, and profile likelihood

The problems in this chapter follow the corresponding topics in the supplied syllabus notes. Each topic has three independent questions designed to test both derivation and application.

7.1 Why hierarchical models are needed

Problem 115. Ten hospitals each have a small sample and their observed success proportions vary widely. Specify a three-level hierarchical model with patient-level outcomes, hospital-specific probabilities, and population-level hyperparameters. Write the joint posterior kernel and explain how information flows from patient observations to hospitals and then across hospitals. Compare the model with no pooling and complete pooling, identifying the inferential cost of each extreme.

Problem 116. A national programme compares treatment effects across districts. Some districts have 5 observations and others have 500. Explain why estimating every district independently can be unstable and why forcing a single common effect can be unrealistic. Formulate a partial-pooling model and predict which districts will be shrunk more strongly toward the population distribution. State which assumptions justify borrowing information across districts.

Problem 117. A grouped dataset is analysed with a hierarchical model, but one group appears to follow a very different data-generating process from all others. Explain why exchangeability is a modelling assumption rather than a guaranteed property of grouped data. Propose diagnostics or model changes that could accommodate systematic group differences without abandoning the idea of shared population structure.

7.2 Exchangeability and partial pooling

Problem 118. For group j, suppose $\bar{y}_j \mid \theta_j \sim N(\theta_j, s_j^2)$ and $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$. Derive the posterior mean and variance of θ_j conditional on (μ, τ^2). Calculate the shrinkage weight for two groups with the same observed mean but sampling variances 1 and 9. Explain why the less precise group is pulled more strongly toward μ.

Problem 119. A set of group labels is randomly permuted, and the hierarchical prior assigns the same joint probability before and after relabelling. Explain why this is exchangeability and why it does not imply that all group parameters are equal. Compare no pooling, complete pooling, and partial pooling in terms of the assumptions each makes and the information each uses.

Problem 120. Suppose the between-group variance τ^2 is estimated to be very small, but one group has an extreme estimate with a small standard error. Determine how the Normal-Normal model treats this group. Discuss whether strong shrinkage is automatically appropriate, how a non-exchangeable outlier could affect the population fit, and what model checking would reveal.

7.3 The Normal-Normal hierarchical model

Problem 121. Three studies report $\bar{y}_j$ with known sampling variances s_j^2. Conditional on (μ, τ^2), assume $\bar{y}_j \mid \theta_j \sim N(\theta_j, s_j^2)$ and $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$. Derive the conditional posterior distribution of each θ_j. Then derive its precision-weighted posterior mean and explain how the weights change as s_j^2 and τ^2 vary.

Problem 122. For a Normal-Normal hierarchy, analyse the limits $\tau^2 \to 0$ and $\tau^2 \to \infty$. Show how the posterior estimates approach complete pooling and no pooling, respectively. Explain why these are limiting cases of one model rather than unrelated estimators, and discuss what the observed group variability tells us about the plausibility of each limit.

Problem 123. Two groups have the same observed estimate, but Group A has standard error 0.1 and Group B has standard error 2.0. Under a common Normal population distribution, derive the two posterior means and variances conditional on the same (μ, τ^2). Explain why equal observed estimates need not lead to equal posterior uncertainty and why the data precision, not just the point estimate, determines shrinkage.

7.4 Marginal likelihood for hyperparameters

Problem 124. Starting from $\bar{y}_j \mid \theta_j \sim N(\theta_j, s_j^2)$ and $\theta_j \mid \mu, \tau^2 \sim N(\mu, \tau^2)$, integrate out θ_j and derive the marginal distribution of $\bar{y}_j \mid \mu, \tau^2$. Write the marginal likelihood across all groups. Explain why this reduces the dimension of the problem and how it can be used to learn the population mean and between-group variance.

Problem 125. A researcher maximises the marginal likelihood for (μ, τ^2) after integrating out all group effects. Another researcher conditions on the group effects and maximises the joint likelihood. Explain why the two optimisations target different likelihood functions. Derive the marginal likelihood in the Normal-Normal model and discuss how integrating out group parameters changes the information available about the hyperparameters.

Problem 126. A colleague says that integrating out a group parameter and maximising over it are equivalent because both remove a nuisance quantity. Use a simple normal hierarchical model to compare marginalisation and profiling. Identify what distribution is averaged over in the former and what optimisation is performed in the latter. Give a concrete circumstance in which the resulting functions of the hyperparameters differ.

7.5 Empirical Bayes

Problem 127. Suppose observed group estimates follow $\bar{y}_j \mid \mu, \tau^2 \sim N(\mu, s_j^2 + \tau^2)$, for $j = 1, \ldots, J$. Derive a method-of-moments estimator for τ^2 in the balanced case with common known sampling variance. Use the estimator in the conditional posterior of a group effect and explain how the estimated between-group variance controls shrinkage. Discuss what happens when the moment estimate is negative and why a boundary estimate of zero corresponds to complete pooling.

Problem 128. An empirical-Bayes analysis first integrates out group effects, maximises the marginal likelihood for hyperparameters, and then plugs the estimates into each group posterior. Write the steps mathematically for the Normal-Normal model. Explain the computational advantage and identify the uncertainty that is ignored when hyperparameters are treated as fixed at their estimates.

Problem 129. Only four groups are available, and the estimated between-group variance is close to zero. An empirical-Bayes report gives very narrow intervals for each group effect. Explain why the small number of groups makes hyperparameter uncertainty particularly important. Propose a comparison with full hierarchical Bayes and discuss how plugging in a point estimate can understate posterior uncertainty.

7.6 Profile likelihood: the syllabus connection

Problem 130. For the Normal-Normal group-summary model, suppose $\bar{y}_j \mid \mu, \tau^2 \sim N(\mu, s_j^2 + \tau^2)$, with known $s_j^2 = s^2$. For each fixed $\tau^2 \geq 0$, maximise the likelihood over μ, substitute the maximiser back into the likelihood, and derive the profile likelihood for τ^2. In the balanced case, identify the maximum-likelihood estimate of τ^2 and explain why the divisor is J, not $J - 1$, in the likelihood maximisation result.

Problem 131. Let ψ be a parameter of interest and λ a nuisance parameter. Define the profile likelihood and compare it with a Bayesian marginal likelihood. Use a normal model with unknown mean and variance to show explicitly which operation is performed on the nuisance parameter in each method. Explain why profiling does not average over uncertainty in λ, whereas marginalisation does.

Problem 132. A report presents a profile likelihood curve for a variance component and interprets the height at each value as the posterior probability of that variance. Explain why this interpretation is incorrect. Describe how a profile likelihood can be used to study the likelihood support for a parameter of interest, what additional ingredients a Bayesian posterior requires, and why the two curves need not have the same shape.

7.7 Full hierarchical Bayes versus empirical Bayes

Problem 133. Write the posterior for a group parameter θ_j under full hierarchical Bayes by integrating its conditional posterior over the posterior distribution of hyperparameters. Write the empirical-Bayes approximation obtained by plugging in $\widehat{\eta}_{EB}$. Explain which uncertainty is propagated in each approach and construct a setting with few groups in which the two resulting intervals differ substantially.

Problem 134. A full Bayesian analysis assigns a hyperprior to (μ, τ^2), while an empirical-Bayes analysis estimates these quantities from the marginal likelihood. Compare the information each method uses and the computational burden of each. Explain why full Bayes is not automatically immune to poor prior or likelihood choices, and why empirical Bayes is not necessarily invalid merely because it plugs in an estimate.

Problem 135. A collaborator argues that empirical Bayes, full hierarchical Bayes, and profile likelihood are three names for the same operation of 'removing nuisance parameters'. Construct a mathematical comparison showing (i) integration over hyperparameters, (ii) plug-in estimation of hyperparameters, and (iii) maximisation over nuisance parameters. Explain the different inferential targets and identify which uncertainty is retained or discarded in each procedure.